OIL LEAK AND FIRE HAZARD

STOP IF OIL IS REACHING HOT EXHAUST OR AN OXYGEN SENSOR

Oil on the manifold is a fire hazard

Engine oil on hot cast iron, on a downpipe, or on an oxygen sensor can ignite. A manifold that has been glowing makes that ignition likely, not theoretical. This problem is separate from the idle code and separate from P0161. It can destroy the engine and the vehicle on its own.

If you see smoke or flame from the oil

Engine off. Everyone out and away. If there is flame, do not open the hood into a fireball if the fire is already rolling out of it. Call the fire department. Use an extinguisher only if you can do it from a safe direction and you have a clear exit. Do not drive the truck away from the fire.

After it is cool — find the fresh leak

1. Confirm the iron is cold enough to hold a hand on. A manifold that glowed needs a long cool-down. Do not degrease a hot pipe. Solvent on hot iron is another fire.
2. Degrease the passenger side, where this engine's manifold and oil-filter adapter live. Clean the valve-cover rail, the filter area, the block side, the sensor connectors, and the top of the pipe. Brake cleaner or a purpose-made degreaser, cold engine, ventilation, no smoking, rags out of the bay and into a metal can. Do not soak electrical connectors. Wipe them.
3. Start only when the pipe is clean and has not been glowing. Idle. Watch for the first wet line of new oil. That line is the source. Old black film lies about direction. Fresh oil does not. UV dye is the right tool if the leak is a seep rather than a run.
4. Repair that source before any oxygen sensor that was sitting in the oil is replaced. A new sensor in the same spray fails again, and the fire hazard is still there.

P0161 does not diagnose the leak, and the leak does not diagnose P0161. Oil in the connector can corrode the heater terminals and cause the code. **POSSIBLE** Oil on the sensor body more often kills the signal than the heater. Test the heater circuit in Guide 02 after the area is clean and dry. Replace the sensor if the test says the heater or the connector is finished. Leave it alone if the circuit tests good.

OIL LEAK · COMMON 4.0L SOURCES

Where a 4.0L usually leaks

Every item below is a common source on this engine. LIKELY as a place to look. None is confirmed until the fresh oil is traced to it. Order the gasket or seal by VIN after you know which one. No part numbers are printed here, so the wrong gasket does not get bought from this page.

Source	Where the fresh oil shows	Trap to avoid
Valve-cover gasket	Passenger-side rail and the corners. Oil runs down the cover and onto the manifold. Often the leak that wets the exhaust highest up. Very common. LIKELY	The cover bolts loosen. Check torque and the bolt seals before you condemn only the gasket. A CCV elbow leaking oil is next to this and is a different part.
Oil-filter adapter O-rings	The adapter between the filter and the block. Oil runs down the block, onto the exhaust, and toward the starter. A classic 4.0L leak. LIKELY	This is often miscalled a rear main. If the oil is at the filter housing, do the adapter. The job is the O-rings, and the surface has to be clean.
Oil-pressure sending unit	The sender threads into the block. A weep runs down from the hex and can look like the adapter or the pan.	Watch whether the wet starts at the sender threads. Replace the sender if it is the source. Do not add thread tape that can enter the oil passage. Follow the sender's seal design.
Oil-pan gasket	Corners and the side rails. Drips lower than the manifold. Road speed can blow it back onto the pipe.	Clean first. A pan leak and an adapter leak can both be present. The higher wet point is the one hitting the manifold at idle.
Rear main seal	Between the engine and the transmission, seen at the inspection cover or as oil thrown around the bellhousing.	Do not replace this seal until the adapter, the pan rail, and the sender have been ruled out with a clean-and-watch. It is the expensive leak, and it is not the most common one on this engine.

Repair the leak before the sensor

When oil has been on an oxygen sensor or in its connector, the leak repair comes first. Then dry the connector, then run the Guide 02 tests. Replace the sensor only if those tests fail, or if the sensor shell and the connector are physically contaminated and cannot be made reliable. A heater that ohms correctly and has power and control does not get replaced "because there was oil nearby."

Prove the repair

After the gasket or seal is done, degrease again. Idle. Look. Then a short drive only if the manifold has stayed dark and no new oil is tracking toward the pipe. Check again when you park. A leak that only appears at speed still needs the UV dye. Recheck the sensor connector for fresh oil before you clear P0161.

Passenger-side note: the 4.0L exhaust manifold and the oil-filter adapter are on the passenger side of this chassis. Confirm that on the truck. A leak on the other side can still blow oil onto a pipe, but the first look is the manifold side.